The National Football League Combine 40-yd Dash: How Important is Maximum Velocity?

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Abstract

Clark, KP, Rieger, RH, Bruno, RF, and Stearne, DJ. The NFL combine 40-yard dash: how important is maximum velocity? *J Strength Cond Res* 33(6): 1542–1550, 2019—This investigation analyzed the sprint velocity profiles for athletes who completed the 40-yard (36.6 m) dash at the 2016 National Football League (NFL) Combine. The purpose was to evaluate the relationship between maximum velocity and sprint performance, and to compare acceleration patterns for fast and slow athletes. Using freely available online sources, data were collected for body mass and sprint performance (36.6 m time with split intervals at 9.1 and 18.3 m). For each athlete, split times were used to generate modeled curves of distance vs. time, velocity vs. time, and velocity vs. distance using a monoexponential equation. Model parameters were used to quantify acceleration patterns as the ratio of maximum velocity to maximum acceleration (v_{max}/a_{max}, or τ). Linear regression was used to evaluate the relationship between maximum velocity and sprint performance for the entire sample. In addition, athletes were categorized into fast and slow groups based on maximum velocity, with independent *t*-tests and effect size statistics used to evaluate between-group differences in sprint performance and acceleration patterns. Results indicated that maximum velocity was strongly correlated with sprint performance across 9.1, 18.3, and 36.6 m (r^2 of 0.72, 0.83, and 0.94, respectively). However, both fast and slow groups accelerated in a similar pattern relative to maximum velocity ($\tau = 0.768 \pm 0.068$ seconds for the fast group and $\tau = 0.773 \pm 0.070$ seconds for the slow group). We conclude that maximum velocity is of critical importance to 36.6 m time, and inclusion of more maximum velocity training may be warranted for athletes preparing for the NFL Combine.

Key Words running performance, sprinting, acceleration, biomechanics

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Introduction

Linear speed is among the most important predictors of success for athletes in many team sports. Previous research has demonstrated that linear speed may distinguish between athletes of different performance levels and position groups in American football (31), soccer (9,33), and rugby union (27). Accordingly, many professional sports leagues test linear sprint performance at predraft scouting combines, including those run by the National Football League (NFL) (29,31), National Basketball Association (20), and Australian Rules Football League (5). For many athletes participating in the NFL Combine, the 40-yard (36.6 m) dash may be the most heavily weighed performance test, as 36.6 m time has been shown to distinguish between drafted and nondrafted players across different position groups (31).

The emphasis on speed during competition and in testing environments has led to numerous investigations into linear sprinting mechanics. For field sport athletes such as collegiate American football players, the 36.6 m dash is commonly completed with split interval times at 9.1 and 18.3 m (2,29,31). Graphs of velocity vs. distance (2) illustrate that the 36.6 m sprint can be divided into an acceleration phase (start to 18.3 m) and a maximum velocity phase (18.3–36.6 m). Previous studies have shown that maximum velocity has a moderate to strong correlation with acceleration performance ($r = 0.56$–0.87) (3,18,21,34) and a strong correlation with overall performance in sprints of 100 m or less ($r = 0.74$–0.95) (25,28,29,32,34). However, presently, the mechanistic link between performance in the acceleration phase and maximum velocity phase is not clearly understood.

In addition to correlational analyses examining acceleration and maximum velocity, several recent studies (10,12,22,30) have examined sprint velocity profiles using a version of the monoexponential equation originally proposed by Hill et al. (13) and more recently popularized by Greene (14), and Chelly and Denis (6). This equation models a runner's velocity vs. time curve as follows:

$$v(t) = v_{max}\left(1 - e^{-\frac{t}{\tau}}\right) \quad (1)$$

where v_{max} is the runner's maximal velocity during the sprint and τ is the time constant. Mathematically,

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τ represents the ratio of v_{max} to initial acceleration (a_{max}) (Figure 1A) or the ratio of v_{max} to initial mass-specific horizontal force (F_H/m):

$$\tau = \frac{v_{max}}{a_{max}} = \frac{v_{max}}{F_H/m} \quad (2)$$

Conceptually, τ represents how quickly a runner accelerates relative to maximum velocity. A small τ value indicates that a runner quickly accelerates to a high percentage of v_{max}, whereas a large τ value indicates that it takes a runner relatively longer to reach a high percentage of v_{max} (Figure 1B).

Recent articles have provided initial evidence that τ values (calculated from mean values of v_{max} and F_H/m) fall within a narrow range for a diverse group of runners, including male field sport athletes and male and female competitive sprinters (10,22,23,28,32). However, an explicit analysis of τ values across runners of different v_{max} has not yet been completed, and a more detailed investigation is intriguing for two major reasons. First, it may provide insight into the underlying link between performance in the acceleration phase and maximum velocity phase because similar τ values indicate that athletes of different sprinting capability accelerate in a comparable manner relative to their individual v_{max}. Second, there are practical applications for coaches and athletes in that, although runners may sprint at maximum velocity for less than half of the 36.6 m test, similar τ values across runners could indicate that v_{max} is more important to short sprint performance than previously recognized.

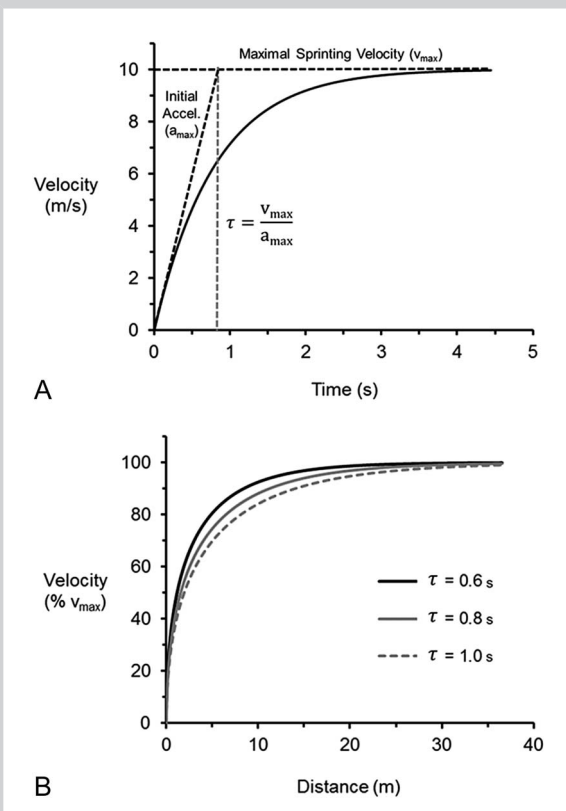

Figure 1. A) Graph of velocity vs. time and calculation of τ value for a hypothetical runner with a v_{max} of 10.0 m·s^{-1} and τ value of 0.8 seconds (modified from Chelly and Denis [6]). B) Graph of % v_{max} vs. distance for 3 hypothetical runners with τ values of 0.6, 0.8, and 1.0 seconds. The smaller the τ value, the faster the runner accelerates relative to v_{max}. Adaptations are themselves works protected by copyright. So in order to publish this adaptation, authorization must be obtained both from the owner of the copyright in the original work and from the owner of copyright in the translation or adaptation.

Therefore, the aim of this investigation was to analyze publicly available data from the 2016 NFL Combine with the intent of evaluating the velocity profiles of all athletes who completed the 36.6 m sprint test. Specifically, the research purpose was to examine the relationship between v_{max} and sprint performance across all split intervals (9.1, 18.3, and 36.6 m times), and to evaluate the acceleration patterns for athletes of different masses and maximum velocities. We hypothesized that v_{max} would be strongly correlated with sprint performance across all 3 split intervals, but that τ values would not be significantly different between fast and slow runners. This would indicate that NFL Combine athletes of varying sprinting capability accelerate in a similar pattern relative to maximum velocity.

Methods

Experimental Approach to the Problem

To test the research questions and hypotheses, the 2016 NFL Combine served as an ideal data source because test metrics were collected on a large sample size from a heterogeneous subject population that varied greatly in anthropometric characteristics and sprinting speed. These combine data were acquired from freely available online sources, with specific focus on metrics of body mass and 36.6 m sprint time (including split times at 9.1 and 18.3 m). Using these 3 split times, curve parameters for each athlete's distance vs. time, velocity vs. time, and velocity vs. distance profiles were determined from a mathematical model based on Equation 1. The curve parameters, τ and v_{max}, were statistically analyzed to test the hypotheses.

Subjects

Data were collected and cross-checked from three websites: NFLdraftscout.com, nflcombineresults.com, and nfl.com. Players were only included in the analysis if measurements of body mass and sprint performance (36.6 m sprint time with split intervals at 9.1 and 18.3 m) were available. Of the 267 players who had data listed for the above metrics, 7 players were not included in the final analysis because of minor discrepancies in sprint times listed across the three websites. Otherwise, data for the

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Table 1. Results for the measured vs. model sprint times (mean ± *SD*).

	Measured times (s)	Modeled times (s)	Absolute error (s)	Absolute % error	Pearson's *r* (correlation coefficient)
9.1 m time	1.673 ± 0.098	1.696 ± 0.095	0.024 ± 0.012	1.4 ± 0.7	0.990
18.3 m time	2.796 ± 0.159	2.773 ± 0.160	0.024 ± 0.011	0.9 ± 0.4	0.997
36.6 m time	4.812 ± 0.296	4.818 ± 0.295	0.006 ± 0.003	0.1 ± 0.1	0.999

remaining 260 players were included in the analysis and deemed accurate. Institutional review board approval was considered a nonissue because this investigation was retrospective in nature, and the data were obtained from freely available online sources.

Procedures

At the NFL Combine, athletes begin the 36.6 m sprint from a 3-point stance, with one hand positioned on the ground at the starting line. Once the athlete reaches a set position in the stance, a momentary pause is required before the athlete can initiate the start at his own volition. The fully automatic timing system (Zybek Sports, Broomfield, CO, USA) is then triggered to start the clock when the athlete's hand lifts off the ground at the start. Split intervals are recorded as the athlete sprints at maximal effort through sensors at 9.1, 18.3, and 36.6 m.

Quantitatively evaluating the research question and hypotheses for this study required generating distance vs. time, velocity vs. time, and velocity vs. distance profiles for each athlete based on the measured sprint times from the 2016 NFL Combine. Profiles of distance vs. time *x(t)* were obtained by integrating *v(t)* from Equation 1:

$$x(t) = \int v(t)\ dt = v_{max}t - v_{max}\tau\left(1 - e^{-\frac{t}{\tau}}\right) \qquad (3)$$

The parameters v_{max} and τ were determined using least-squares regression to obtain a best-fit curve for each athlete's distance vs. time profile based on the listed split times from 9.1, 18.3, and 36.6 m.

In addition, Equations 1 and 3 were used to generate velocity vs. distance profiles for each athlete. Velocity was quantified both in absolute units (velocity in $m \cdot s^{-1}$) and in relative units (percentage v_{max}). Generating profiles of percentage v_{max} vs. distance in relative units provided a qualitative, visual analysis of how athletes of different v_{max} accelerate relative to maximum velocity. Finally, percentage v_{max} at specific distance intervals (5, 10, 15, and 20 yards; or 4.6, 9.1, 13.7, and 18.3 m) was calculated to provide an interpretation of the percentage v_{max} vs. distance graphs and a secondary analysis of whether fast vs. slow accelerate in a similar pattern relative to v_{max}.

Statistical Analyses

The accuracy of the measured vs. modeled split times at 9.1, 18.3, and 36.6 m was evaluated using absolute error, absolute percentage error, and Pearson's *r* correlation

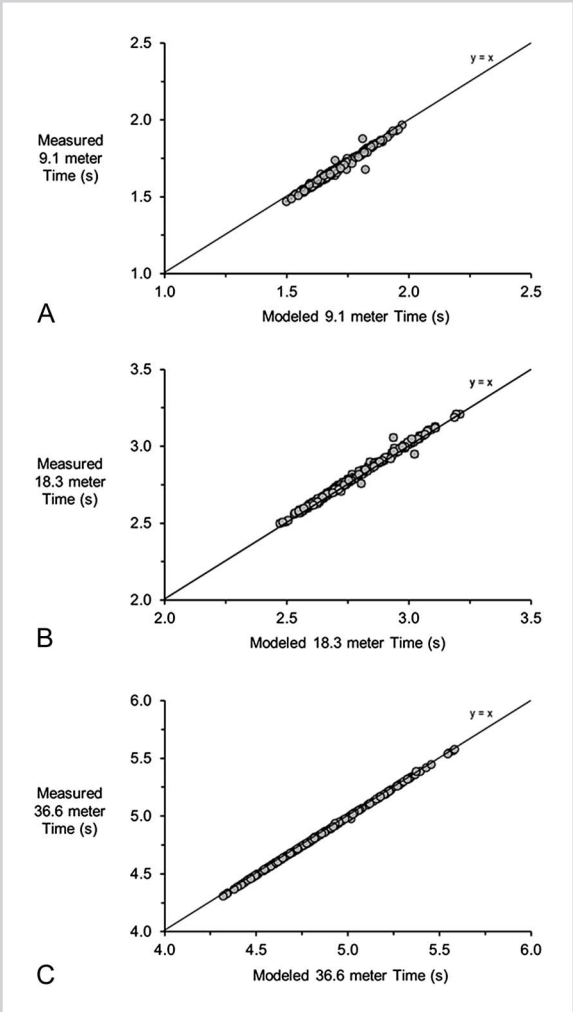

Figure 2. Measured vs. modeled times for (A) 9.1 m, (B) 18.3 m, and (C) 36.6 m splits for all athletes in the database.

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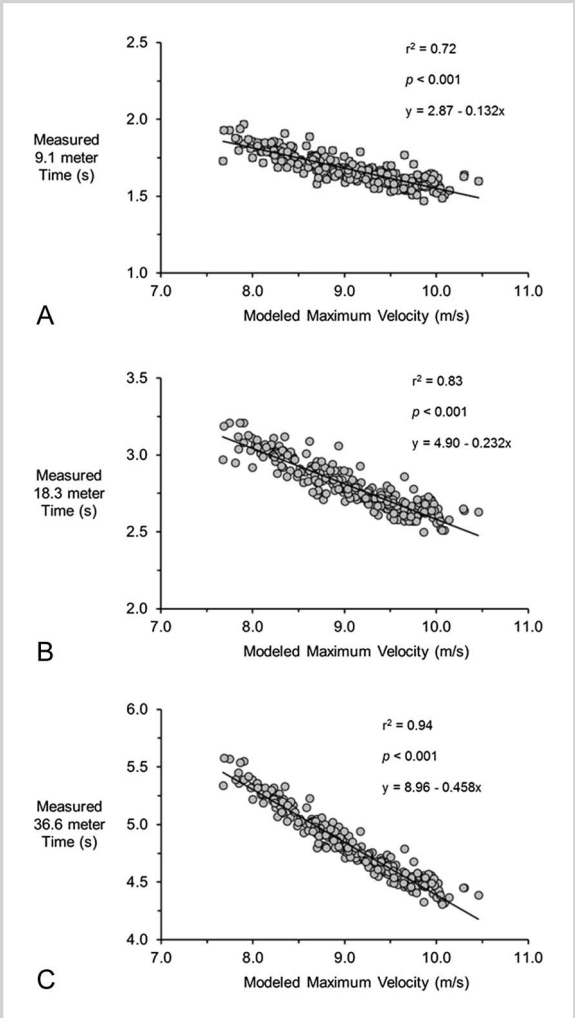

Figure 3. Relationship between v_{max} and sprint performance for (A) 9.1 m, (B) 18.3 m, and (C) 36.6 m splits for all athletes in the database.

coefficient. Absolute percentage error was calculated as follows:

$$\text{absolute percentage error} = \left| \frac{\text{measured} - \text{modeled}}{\text{measured}} \right| \times 100$$

To examine the relationship between v_{max} and overall sprint performance, the subjects were analyzed as an entire group ($n = 260$). The correlation between v_{max} and sprint performance across all split intervals was evaluated using simple linear regression to generate a best-fit equation and coefficient of determination (r^2), using the significance level of 0.05. To examine whether τ values varied between runners of different speeds, the subject pool was divided into fast ($n = 130$) and slow ($n = 130$) groups based on v_{max}. Between-group differences for mass, split intervals (9.1, 18.3, and 36.6 m times), v_{max}, and τ values were evaluated with separate independent *t*-tests on each variable, using a statistical significance level of 0.05. In addition, effect size statistics (Cohen's *d* (8)) were used to determine the magnitude of difference between fast and slow groups, with *d* calculated as the difference between group mean values divided by the pooled *SD*. Threshold values for *d* statistics were as follows: <0.20 (trivial), 0.20–0.60 (small), 0.61–1.20 (moderate), 1.21–2.00 (large), and 2.01–4.00 (very large) (16). All statistical analyses were completed using Microsoft Excel and SAS (Statistical Analysis Software, USA).

Results

Differences between the measured and modeled times for each athlete were analyzed for 9.1, 18.3, and 36.6 (Table 1 and Figures 2A–C). For measured vs. modeled times across each of the 3 split intervals, the mean absolute error was less than 0.03 seconds, and the mean absolute percentage error was less than 1.5%. In addition, the correlation between measured and modeled times was $r \geq 0.990$ across each of the 3 split intervals.

When analyzed for the sample as a whole, v_{max} displayed a strong, negative correlation with sprint

Table 2. Results for mass, sprint performance, v_{max}, and τ values for all subjects and grouped by fast vs. slow athletes (mean ± *SD*).*

	Measured data from 2016 National Football League combine				Modeled data	
	Mass (kg)	9.1 m time (s)	18.3 m time (s)	36.6 m time (s)	v_{max} (m·s^{-1})	τ (s)
All	112.3 ± 20.5	1.67 ± 0.10	2.80 ± 0.16	4.81 ± 0.30	9.08 ± 0.62	0.770 ± 0.069
Fast v_{max}	96.7 ± 10.0	1.61 ± 0.05	2.67 ± 0.07	4.58 ± 0.12	9.59 ± 0.30	0.768 ± 0.068
Slow v_{max}	128.0 ± 15.9†	1.74 ± 0.08†	2.92 ± 0.12†	5.05 ± 0.22†	8.56 ± 0.39†	0.773 ± 0.070
Fast vs. Slow *d*	2.36	1.96	2.44	2.69	2.99	0.08
(Interpretation)	(very large)	(large)	(very large)	(very large)	(very large)	(trivial)

*Effect size (*d*) is listed with interpretation.
†Fast vs. slow between-group differences significant at $p < 0.001$.

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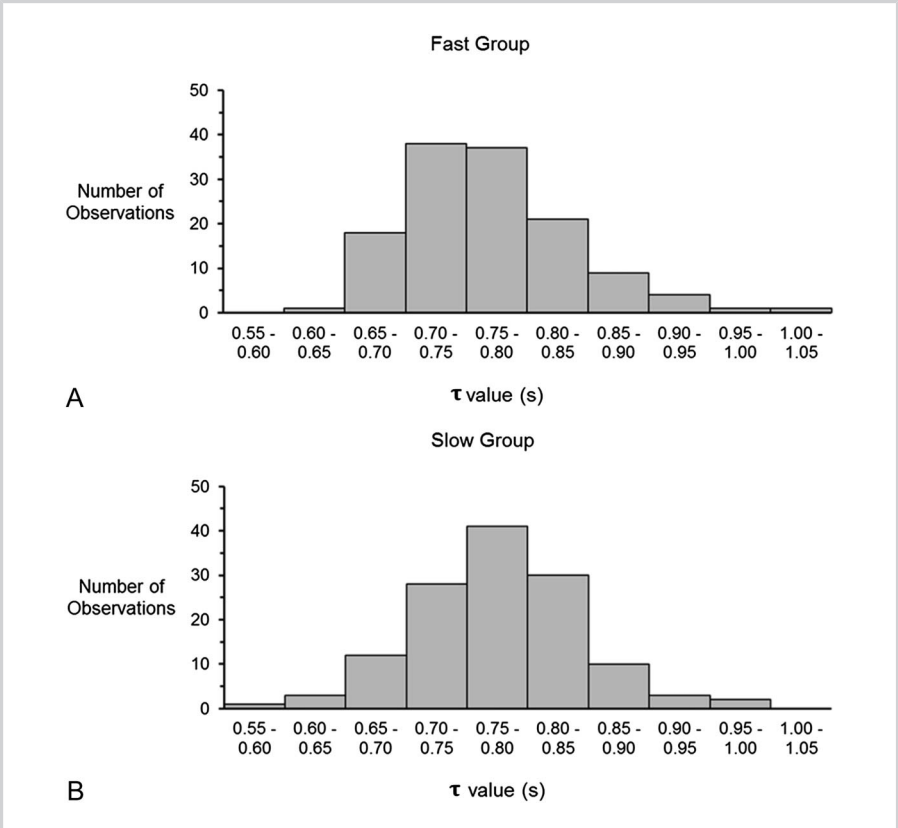

Figure 4. Histogram of τ values for (A) athletes in the fast v_{max} group ($n = 130$), and (B) athletes In the slow v_{max} group ($n = 130$).

performance across each of the measured sprint intervals (Figures 3A–C). Coefficient of determination (r^2) values for v_{max} and sprint times were 0.72, 0.83, and 0.94 for the distances of 9.1, 18.3, and 36.6 m, respectively. In addition, although not depicted in Figure 3, v_{max} displayed a strong, negative correlation with 9.1–18.3 m split time ($p < 0.001$, $r^2 = 0.82$) and 18.3–36.6 m split time ($p < 0.001$, $r^2 = 0.97$).

A summary of the measured and modeled data, including between-group differences for fast and slow athletes, is presented in Table 2. A series of independent t-tests revealed that there were statistically significant between-group differences for the variables of mass, sprint times (at 9.1, 18.3, and 36.6 m), and v_{max}, but no statistically significant between-group differences in τ values ($p = 0.52$). In addition, the magnitude of differences between fast and slow groups was large or very large ($d \geq 1.96$) for the variables of mass, sprint times (at 9.1, 18.3, and 36.6 m), and v_{max}. However, the magnitude of difference between fast and slow groups for τ values was trivial ($d = 0.08$). Finally, although τ values ranged from 0.63 to 1.01 seconds for the fast group and 0.58–1.00 seconds for the slow group, 225 of the 260 total τ values fell within the range of 0.65–0.85 seconds (Figure 4).

The percentage v_{max} at specific distance intervals (4.6, 9.1, 13.7, and 18.3 m) during the acceleration phase was generally similar between fast and slow groups (Table 3). Although the slow group achieved a slightly higher percentage v_{max} at each of these distances, there was less than 2.5% difference between the groups. This is illustrated in Figure 5, which displays how 6 representative athletes from the 2016 NFL Combine (the slowest athlete, athletes representing the 20th, 40th, 60th, and 80th percentile, and the fastest athlete) have very different v_{max} and 36.6 m sprint performance, but similar acceleration patterns when velocity vs. distance is graphed as percentage v_{max}.

Table 3. Results for % v_{max} for all subjects and grouped by fast vs. slow athletes (mean ± *SD*).

	% v_{max} at 4.6 m	% v_{max} at 9.1 m	% v_{max} at 13.7 m	% v_{max} at 18.3 m
All	75.7 ± 2.4	88.9 ± 1.8	94.5 ± 1.3	97.2 ± 0.9
Fast v_{max}	74.5 ± 2.1	88.0 ± 1.7	93.9 ± 1.2	96.8 ± 0.8
Slow v_{max}	76.8 ± 2.0	89.8 ± 1.5	95.1 ± 1.0	97.6 ± 0.6
Fast vs. slow Δ% v_{max}	2.3	1.8	1.2	0.8

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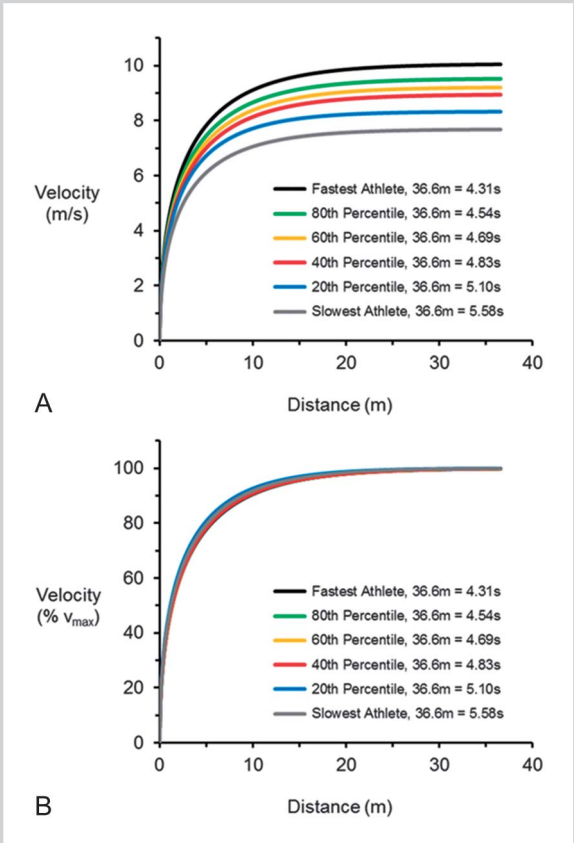

Figure 5. A) Graph of velocity vs. distance for the slowest athlete at the combine, athletes representing the 20th, 40th, 60th, and 80th percentile, and the fastest athlete at the combine. These 6 athletes displayed appreciably different v_{max} and overall 36.6 m sprint performance. B) Graph of % v_{max} vs. distance for the same 6 athletes. The overlapping curves between subjects indicate similar τ values, demonstrating that these athletes of vastly different sprinting capability accelerated in a comparable pattern relative to v_{max}.

Discussion

We performed this analysis with the goals of examining the relationship between maximum velocity (v_{max}) and performance in the NFL Combine 40-yard (36.6 m) dash, and to evaluate whether athletes of different sizes and sprinting capabilities have comparable acceleration patterns. Using available data for 36.6 m sprint performance with split times at 9.1 and 18.3 m, sprint velocity profiles were mathematically modeled for 260 participants from the 2016 NFL Combine. We hypothesized that v_{max} would be strongly correlated with performance in all phases of the sprint, but that both fast and slow runners would accelerate in a similar manner relative to v_{max} (quantified by τ values). Results conclusively supported our first hypothesis, as v_{max} was strongly and negatively correlated with split times during the acceleration phase (9.1 and 18.3 m) and with overall 36.6 m performance. However, no statistically significant differences were found in τ values between fast and slow groups, and only a trivial magnitude of difference between groups. Therefore, we conclude that v_{max} is of critical importance in the 40-yard dash, and that most NFL Combine athletes accelerate in a similar manner relative to maximum velocity.

The close agreement in measured vs. modeled times (Table 1 and Figures 2A–C) indicates that the mathematical procedures were successful in creating accurate distance vs. time, velocity vs. time, and velocity vs. distance profiles for the 260 athletes analyzed. The accuracy of this approach was not guaranteed a priori. Another recent study completed similar velocity profiling using numerous split times to model 100 m dash performance (32), but to our knowledge, this study was the first to investigate velocity profiles of sprints ≤40 m using only 3 split times. Furthermore, while the use of radar guns has recently become popular as a method for collecting instantaneous velocity vs. time data and its derivatives (3,6,10,12,22–25,30), the results from this study provide evidence that this type of velocity profiling may be successfully completed for short sprints using only 3 split times.

Although the close agreement of measured vs. modeled times demonstrates the accuracy of the mathematical procedures and provides quantitative support for the experimental approach, it is worth noting that the τ values generated for the present investigation are smaller than τ values in previously published studies. In this analysis, the mean τ value for the entire study sample was 0.770 ± 0.069 seconds, with similar mean values for the fast and slow groups (Table 2). In previous research, the τ values (calculated from group mean values of v_{max} and F_H/m) typically fall within a narrow range of 1.15 ± 0.15 seconds (10,22,23,28,32). The lower τ values in this study are probably due to the timing mechanism used at the start of the 36.6 m sprint at the NFL Combine. As previously described, the fully automatic timing mechanism is initiated when the athlete's base hand lifts off the ground. However, because force production with the feet pushing backwards against the ground likely commences slightly before the hand leaves the ground (30), there may exist a slight temporal offset between the initial forward acceleration of the center of mass and the trigger of the timing mechanism. This may cause the τ values generated from our mathematical modeling procedures to be slightly lower than the calculated τ values from previous publications that have used radar data and inverse dynamics. However, because the temporal offset between-ground force application and the hand lifting off the ground were presumably similar among all subjects in this study, we did not view this as a confounding variable in our analysis.

Although v_{max} was most strongly correlated with overall 36.6 m time, it was also highly correlated with performance in the acceleration phase (9.1 and 18.3 m split times) (Figures 3A–C). These results are consistent with previous research that has demonstrated a strong relationship between maximum velocity and overall performance in short sprints

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(25,28,29,32,34). Furthermore, although this is certainly not the first investigation to demonstrate the correlation between v_{max} and performance in the acceleration phase, the correlations were slightly stronger in this study than in several previous reports (3,18,21). In addition, although the relationship between performance in the acceleration phase and maximum velocity phase at the NFL Combine has been investigated before (29), this is the first quantitative analysis performed since the NFL Combine recently switched timing methodologies to a fully automatic timed start and finish. Before 2011, the start was manually initiated by a human test administrator. Thus, the stronger correlations between v_{max} and acceleration phase performance in the present analysis could be due to timing methods without the potential for human variability in initiating the starting mechanism.

It should be noted that the large correlation between v_{max} and sprint performance observed in this study was undoubtedly due, in part, to the heterogeneous composition of the participants. Statistically, we analyzed the group as a whole, without categorizing players by position or controlling for mass, as has been done before in similar analyses (29). We justified our statistical approach on the basis of purposely evaluating the macro-level relationship between v_{max} and sprint performance, which required including athletes of vastly different masses in the analysis. In addition, correlation in this case does not necessarily imply causation, and the strong relationship between v_{max} and 9.1 m/18.3 m time does not necessarily indicate that maximum velocity determines acceleration performance. Clearly, if an athlete does not execute the start and acceleration portion of the sprint with a requisite amount of motor skill, overall sprint performance will suffer, regardless of the athlete's maximum velocity. However, given that most runners accelerated in a similar manner relative to maximum velocity in this study, it may be that v_{max} serves as an upper threshold or limiting factor to performance in the acceleration phase. Maximum velocity acting as an upper threshold for performance certainly has precedent in other areas of sprinting research (4).

Our secondary hypothesis was also supported by the results. There were no statistically significant differences in τ values between fast and slow groups, and the magnitude of differences between fast and slow groups was trivial. Although τ values ranged from ~0.6 to 1.0 seconds in both fast and slow groups, a certain amount of variance is likely to be demonstrated in any heterogeneous population, and the large majority of τ values fell in a very narrow range (Figure 4). In addition, the mean percentage v_{max} at specific distance intervals (4.6, 9.1, 13.7, and 18.3 m) during the acceleration phase was similar between groups, with less than 2.5% mean difference at all distances analyzed (Table 3). Therefore, we conclude that both fast and slow groups generally accelerated in a similar manner relative to v_{max}.

Conceivably, the narrow range of τ values observed in this analysis could be due to the fact that all athletes were American football players, and that the acceleration patterns demonstrated in this investigation are specific to this athletic population. Furthermore, many of these athletes likely completed a preparatory program at their university or a private training facility (15) in the 4 to 8 weeks leading up to the NFL Combine. These acute preparatory programs often emphasize combine-specific training regimens, rather than football-specific training regimens. Therefore, although the athletes in this investigation were heterogenous with regard to size and speed, it is possible that the sprint training programs they completed in preparation for the combine were quite homogenous. Future research is needed to determine whether the results from this investigation are generalizable to athletes from a variety of sports and training backgrounds.

Despite the caveats noted above, we believe that the comparable τ values for fast and slow groups could also indicate a broader mechanistic link between acceleration and maximum velocity performance, as highlighted by further examination of Equation 2. Conceptually, τ represents the ratio of v_{max} to initial acceleration (a_{max}), or the ratio of v_{max} to initial mass-specific horizontal force (F_H/m). Newtonian mechanics dictate, and recent experimental research has confirmed, that acceleration performance is related to the amount of mass-specific horizontal force that can be generated and transmitted to the ground (17,28). Furthermore, because the body posture is upright at maximum velocity and nearly all the forces are directed downward, v_{max} is related to mass-specific vertical force (7,35,36). Therefore, the missing conceptual link could simply be that the runners who generate the most mass-specific vertical force (and have a faster v_{max}) are the same runners who generate the most mass-specific horizontal force during initial acceleration. This "mass-specific force" explanation implies that the ratio of v_{max} to a_{max} in Equation 2 remains relatively constant across a heterogeneous population, causing τ values to be generally similar across runners with a diverse range of maximum velocities.

The results of this analysis clearly emphasize the importance of v_{max} for short sprint performance. This does not discount the importance of the start and acceleration phase, as comprehensive speed development programs include training to enhance each of the different phases of a linear sprint (11). Rather, the results of this study simply suggest that it may be worth devoting extensive training time to the maximum velocity phase when preparing for the NFL Combine, even for football players who rarely reach maximum velocity in a game setting. Accordingly, a valuable observation from the mean data at 4.6, 9.1, 13.7, and 18.3 was the high percentage v_{max} that athletes in both groups achieved so early in the sprint (Table 3). Although recent publications have reported velocity vs. distance or velocity vs. time profiles for both sprinters (19,26) and field sport athletes (1–3,24), to our knowledge, this is the first study to analyze sprint profiles in terms of percentage v_{max} vs. distance for

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field sport athletes. Although it is well known that elite sprinters do not approach v_{max} until 50–60 m in a 100 m dash (19,26) and that field sport athletes may reach maximum velocity earlier (1,2), the results of this investigation provide new insight into exactly how fast (in terms of percentage v_{max}) the NFL Combine athletes were running early in the sprint. The results clearly demonstrate that athletes completing the 36.6 m sprint were running at equal to or greater than 90% v_{max} by 13.7 m. This has important practical applications for linear speed development exercise prescription, as anything longer than 13.7 m (15 yards) should perhaps be considered "top speed training" for football players.

In addition, normative data generated in this analysis may be of substantial value to coaches and athletes who are training to improve short sprint performance. As expected, the results indicate that performance in the maximum velocity phase (18.3–36.6 m split time) had a near perfect relationship to modeled v_{max} ($r^2 = 0.97$). Thus, a highly accurate approximation of v_{max} can be calculated by having an athlete complete a flying 18.3 m (20-yard) sprint using automatic timing gates and simply dividing distance by time. Based on the data and regression equations displayed in Figure 3, the calculated v_{max} can be used to determine whether the athlete is accelerating to 9.1 and 18.3 m within the range of expected times based on v_{max}. If these expected times are not being achieved, it may highlight that the velocity curve is not optimal, and modifications to the athlete's mechanics or training program may result in improved performance.

Practical Applications

The results of this analysis indicate that maximum velocity (v_{max}) may be more related to short sprint performance than previously recognized, since groups with fast and slow v_{max} both accelerated in a comparable manner relative to maximum velocity. This is not to discount the importance of the start and acceleration phase, as coaches and athletes clearly must train all portions of a linear sprint. However, the results of this study emphasize that, even for football players who rarely reach peak speeds in a game setting, devoting extensive training time to enhancing maximum velocity may be a critical component when preparing for short sprint tests like the NFL Combine 40-Yard Dash.

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